CIA-1

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2 Given

Energy operation Durabion = 40 min

NO. of cell boundries cell crossed = 15.

Speed of the ambulence = 90 Km/h.

PLAN Handoff Requires = 120ms

BSS coordination Requires = ~~350~~ 80 ms.

PLAN coordination Requires = 120 ms.

BSS path switching = 80ms

PLAN Aditional Authenbrcation = 350 ms

Roming Registration = 280 ms

No. of Times Failed = 3 Times

Aditional Recovery Time = 400ms

Total NO. of Loss = 512 Kbps

Total normal hand off delay :-

$D_{delay} = 120 + 80 = 200ms$

$=$

$D_{total} = 200 \times 15 = 3000ms$

$= 3000ms = 3sec$

the total normal hand off delay is 3sec

Aditional Recovery Delay :-

$$\text{Recovery} = 400 \times 3 = 1200 \text{ ms}$$
$$= 1200 \text{ ms} = 1.2 \text{ sec}$$

Aditional Recovery Delay is 1.2 sec

Roming Relakd delay :-

$$\text{Droming} = 350 + 250 = 600 \text{ ms}$$
$$= 600 \times 2 = 12000 \text{ ms}$$
$$= 12000 \text{ ms} = 1.2 \text{ sec}$$

Voverall Communication intruption :-

$$D_{total} = 3000 + 1200 + 1200$$
$$= 5.4 \text{ sec} = 5400 \text{ ms}$$

System satisfies the intruption limit of 8 sec

$$5.4 > 3 \text{ sec}$$

overall communication exceed the Limit of 2.4 sec

Potential Patent Data Loss :-

$$D_{data} = 512 \text{ KBPS}$$
$$= 512,000 \text{ bits}$$

$$T_{total} = \frac{512,000}{8 \times 1000} =$$

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Potential Palent Data Loss :-

$$D_{data} = 512 \text{ KBPS}$$
$$= 512{,}000 \text{ bits}$$

$$T_{total} = \frac{512{,}000}{8 \times 1000} = \frac{512{,}8000}{8000}$$

$$= 64 \text{ ms}$$

$4888.4 \times 8 \times 1000$

$= 39{,}107{,}200$ Kbits

1 Sol

Hexagonal cellular Network across = 1200 Km
Cell Radius = 1.8 Km
Avaliable Spectrum = 48 MHz
full duxex channel = 200 KHz
Network performance for Frequency Reuse N=7 & 4
Reduction in usable channels = 15 sec
Minimum Reuse Ratio = 4.5

$$\text{Area} = \frac{3\sqrt{3}R^2}{2} = 2.598(1.8)^2$$
$$= 2.6 \times 3.24$$
$$= 8.32$$

i) Total No. of cells $= \frac{1200}{8.32} = 143$

ii) Total avaliable channels $= C = \frac{48 \times 1000}{200}$
$= 240$ Hz

iii) ~~for~~ For $N(7) = \frac{240}{7} = 34.27$

For $N(4) = \frac{240}{4} = 60$

$4 \frac{3}{85 \times 4}$
340

N=7 Reduction $= 34.27 \times 0.85$
$= 29$ channels/ unit

N=4 Reduction $= 60 \times 0.85$
$= 51$ channels/ unit

iv) Frequency Reuse Ratio N=7

$R = \sqrt{3N} = \sqrt{21}$

distance $D = Q \times R$
$= 4.58 \times 1.8$

$D = 8.28$

Fxlency Reuse for $N=4$

$$R = \sqrt{3 \times 4} = \sqrt{12}$$

$$D = Q \times R$$

$$D = 1.8 \times \sqrt{12}$$

$$= 1.8 \times 346$$

$$= 6.24$$

∴ $N=7$ is the usable choice because it provide better inteference. and $N=4$ is not suitable. the Reuse Distance at $N=7$ is 8.28 and $N=4$ is 6.24.

3

i) No. of GSM base station controller

No. of selectors = 3 selector

No. of Transceivers = 4 Transceivers

No. of TDMA provides $TR_x = 2$

Total subscribers = 1200

Total No. of call in 1 hr = 12 Duration of 90 sec

Traffic in each channel = 16 Kbps

BTS connected Required =

Analyze base station sector:-

Time slots per sector :- $4(8) = 32$

$= 32 - 2$

$= 30$

per BTS :- $30(3) = 90$

BSC $= 90(12) = 1080$

3) Given
GSM operator manages = 12 BTS

b) Offered traffic per sector :-

$$A = \frac{1200\,(1.2)\,(90)}{3600}$$

$$A = 36\ \text{E range}$$

c) Channel Utilization :-

$$U = \frac{\text{offered traffic}}{\text{Avai channels}} \times 100$$

$$U = \frac{36}{30} \times 100$$

$$U = 120\ \%$$

d) Min TRXs Req per sector :-

$$80\ \% \Rightarrow A_{max} = 0.8\,(8T-2)$$

$$36 \leq 0.8\,(8T-2)$$

$$45 \leq 8T-2$$

$$47 \leq 8T$$

$$T \geq 5.875$$

$\therefore$ T = 6 TRXs per sector

Utilization $\frac{36}{46} \times 100 \Rightarrow 78\ \%$

3
ii)

$E_b/N_0 = 8$

voice activity factor = 0.4

Frequency efficiency = 0.6

CDMA reuse factor = 1

AT FDM for CDMA = 1.25

TDMA carrier bandwidth = 200KHz

voice data rate = 96 Kbps

a) FDMA capacity

$C_{FDMA} = \frac{1000}{30} = 333.33$

$C_{cell} = \frac{333.33}{7} = 47.62$

total user per cell = 47.62

b) TDMA capacity

C_{TDM} capacity $= \frac{100}{2} = 50$

$C_{cell} = \frac{50}{5} = 10$

$C_{cell} = 10$ //

c) CDMA capacity :-

$C_{CDMA} = \frac{10{,}1000}{1.25} = 7500$

$C_{cell} = \frac{7500}{100} = 75$

d) CDMA User per carrier

$K = \frac{G_P}{(E_b/N_0)}$

$K = \frac{128\ (0.6)}{8\ (0.4)}$

$P_{cell} = 8(24) = 192$

192 user per cell

e) capacity comparision :-

Technique

FDMA

TDMA

CDMA

4

Sol Total No of Transfered Img Files = 240
Each Img size 18 MB
No of time slots = 4 slots
theoritical data Rate = 20 Kbps
Reduce capacity by = 18%
Realc conjuction Limit = 85%.
Packet Transaction Rate increases by 12%
2.5 sec for comparing and analysis
UMS Transfer Time = 384 Kbps

R = No of time slots × Rate per

R = 4×20
R = 80 Kbps

$R_1 = 80 \times (1 - 0.18)$

$R_1 = 80 \times 0.82 = 65.6$ Kbps

$R_2 = 65.6 \times 0.85$

$R_2 = 55.76$ Kbps

GPRS throughput = 55.76 Kbps

D = 240 × 18
D = 4320 MB
$D_T = 4320 \times (1 + 0.12)$
$D_T = 4838.4$ MB

MB to KB
4838.4 × 8 × 1000
= 38,707,200 Kbits

$T = \frac{38,707,200}{55.76}$

T = 694174.97

TI = 8 × 2.5 = 20

$T_G = 694195$

~~TG =~~